



DWARAKA DOSS GOVERDHAN DOSS VAISHNAV COLLEGE
(AUTONOMOUS)
College with Potential for Excellence
Linguistic Minority Institution, Affiliated to University of Madras
PG DEPARTMENT OF COMPUTER SCIENCE



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CIA I – AUG 2026

INFORMATION SECURITY-2527318(c)

II M.Sc (CSC)

Max. Marks : 50

DATE : 19.8.26

SECTION A (10*1=10 marks)

Answer ALL questions

1. Define threat.
2. What is meant by confidentiality?
3. State one difference between authentication and access control.
4. What is fabrication in network security?
5. What is a security mechanism?
6. What is an attack?
7. What is the purpose of a digital signature?
8. Define a digital certificate.
9. What is the role of a Certificate Authority (CA)?
10. What is Kerberos?

SECTION B (4*5=20 marks)

11. a). Explain the different types of active attacks with suitable examples.
Or
b) Describe the different types of passive attacks with suitable examples.
12. a). Explain the principles of public key cryptography with suitable diagram.
Or
b) Explain Diffie Hellman's key exchange algorithm with example.
13. a) Explain the process of digital signature generation and verification.
Or
b) Discuss the structure and purpose of a digital certificate.
14. a) Explain the working of Kerberos authentication protocol.
Or
b) Describe the X.509 Directory Authentication Service and its features.

SECTION C (2*10=20 marks)

15. Explain the RSA algorithm in detail with key generation, encryption and decryption process.
16. a) Explain in detail Security services and mechanisms used to implement these services.
Or
b) Describe the Kerberos authentication system and explain the X.509 framework with suitable diagrams and message flow.



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CIA I – AUGUST 2026
DOT NET TECHNOLOGY

II M.Sc. (CS)

Subject Code: 2527316

Max. Marks: 50

Date: 14.08.2026

SECTION A (10*1=10 marks)
Answer ALL questions

1. What is the .NET Framework?
2. Name any two primitive data types in C#.
3. What is a method in C#?
4. What is an array?
5. What is C#?
6. Which keywords are used to handle exceptions in C#?
7. Define inheritance.
8. What is an event in C#?
9. What is an interface in C#?
10. What is the difference between a class and a structure?

SECTION B (4*5=20 marks)

- 11.a) Explain the C# and its .Net Architecture

OR

- b) Determine the usage of Jagged array using for loops with example

- 12.a) Demonstrate any two branching statements in C# with an example.

OR

- b) Explain arrays, structures and enumeration with an example.

- 13a) Define Delegates and Explain with an example.

OR

- b) Write a Short note on Events with an example.

- 14a) Explain Exception handling using try-catch blocks.

OR

- b). Discuss about Interface with an Example.

SECTION C (2*10=20 marks)

15. Explain types of inheritance with an example program.

- 16.a) Construct a console application demonstrating structures in C# with example

(or)

- (b) Create a console application to implement class and object with necessary explanation and example program.

1. What is the .NET Framework?
2. Name any two primitive data types.



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PG DEPARTMENT OF COMPUTER SCIENCE
I CIA- AUG 2026
BIG DATA ANALYTICS

Class: II M.Sc. (Computer Science)
Course Code : 2527317

Max. Marks: 50
Date : 17.08.2026

SECTION A (10*1=10 marks)

Answer ALL questions

1. Define Big Data?
2. How can big data add value to business?
3. What are the sources of big data?
4. Define web analytics.
5. What is algorithmic trading?
6. Define NOSQL
7. What is the purpose of an aggregate data model?
8. What is versionstamp?
9. Define Key Value Data model.
10. Write down the process of relaxing consistency.

SECTION B (4*5=20 marks)

11. a) Elaborate the Industry example of big data.
(or)
b) Write the Characteristics of Big Data.
12. a) Identify the various types of risk management.
(or)
b) Describe the uses of big data in health care and medicine.
13. a) Explain NoSQL features, advantages, disadvantages.
(or)
b) Write short notes on Schemaless Database.
14. a) Discuss about the various forms of consistency.
(or)
b) Experiment the process of MapReduce in detail.

SECTION C (2*10=20 marks)

15. Explain the advantages of big data in organizations.
16. a). Differentiate: Replication and Sharding Distribution Models.
(or)
b) Explain the NOSQL Data Models.